MIL-R-43954A
11 June 1985
SUPERSEDING
MIL-R-43954
10 November 1976

MILITARY SPECIFICATION

RANGES, CONVECTION OVEN, ELECTRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers an electric range with convection oven(s) and with either griddle top, or combination of griddle and hot plates, or with six hot plates.

1.2 Classification. The ranges shall be of the following types, grades, sizes and styles, as specified (see 6.2).

Type I	-	Range with griddle top
Type II	-	Range with griddle top and hot plates
Type III	-	Range with six hot plates
Grade A	-	Stainless steel - 300 series (non-magnetic)
Grade B	-	Stainless steel - 400 series
Size 1	-	Range - 36 inches wide by 38 inches deep by 30 inches high
Size 2	-	Range - 42 inches wide by 28 inches deep by 46 inches high
Style 1	-	Fully assembled range (ready for in-place installation)
Style 2	-	Modular bolted construction (for assembly in place)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this letter.

FSC 7310

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:10Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-11337 - Ranges, Packaging and Packing of
MIL-H-22577 - Heating Elements, Electrical: Cartridge, Strip and Tubular Type

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

UNDERWRITERS LABORATORIES, INC.

UL 197 - Commercial Electric Cooking Appliances

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

NATIONAL SANITATION FOUNDATION (NSF)

No. 4 - Commercial Cooking and Warming Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, 3475 Plymouth Road, P.O. Box 1468, Ann Arbor, MI 48106.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:10Z
Check the source to verify that this is the current version before use.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02169.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet and Strip
A 176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet and Strip
A 342 - Test for Permeability of Feebly Magnetic Materials
A 436 - Austenitic Ductile Iron Castings
A 463 - Steel Sheet, Cold Rolled, Aluminum Coated Type 1
A 510 - General Requirements For Wire Rod And Coarse Round Wire, Alloy Steel
A 580 - Free-Machining Stainless And Heat-Resisting Steel Wire
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Standard product. The range shall, as a minimum, be in accordance with the requirements of this document and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this document. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this document shall be identified by all regular manufacturers or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

Source: https://assist.dla.mil -- Downloaded: 2026-04-21T23:10Z
Check the source to verify that this is the current version before use.

3.2 Code and standards. The electric convection ranges shall comply with the applicable requirements of UL 197, except as otherwise specified (see 3.2.1.1), and NSF No. 4 and NFPA 70.

3.2.1 Compliance. Prior to approval of the initial production lot, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative, that the electric convection ranges he proposes to furnish meet the applicable requirements of UL 197 and NFPA 70, and NSF No. 4 as follows:

3.2.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL listing or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the style 1 convection ranges have been tested and conform to UL 197. For style 2 ranges modular construction shall conform to UL approved electrical components.

3.2.1.2 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

a. A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished electric convection range, or

b. A certification for the electric convection range issued by NSF under their special one-time contract evaluation/certification service, or

c. A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the electric convection ranges are being procured, indicating that they have been tested and conform No. 4.

3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial electric convection range provided the completed item complies with all provisions of this document (see 6.3).

3.3.1 Cast iron. Grey iron alloy castings shall conform to ASTM A 436.

3.3.2 Stainless steel. Stainless steel shall conform to any type 300 series (except 303 for welding), as specified in ASTM A 167, and any type 400 series (type 405 to be used for welding only), as specified in ASTM A 176. Stainless steel wire shall conform as specified to ASTM A 580.

3.3.3 Aluminized steel. Aluminized steel shall conform to T1-40 commercial coating as specified in ASTM A 463.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:10Z
Check the source to verify that this is the current version before use.

3.3.4 Carbon steel. Carbon steel shall conform to commercial quality as specified in ASTM A 569 or ASTM A 510.

3.3.5 Insulation. Insulation for all ranges shall be of the blanket or block type installed in accordance with the manufacturer's commercial practice. Loose fill type insulation shall not be used.

3.4 Design. The ranges shall be a free standing, heavy duty commercial design with electrically heated griddle top and hot plates and convection ovens. The range ovens shall have blowers for forced air circulation throughout the compartment. Each oven shall have tubular clamp-on Calrod-type heating elements located behind a removable protective baffle plate on the side(s), backwall or top of the oven compartment so as to provide even temperature distribution throughout the oven. The ovens shall be provided with either removable or non-removable shelf supports. Each range shall be equipped with a vent, actuated by a manual control on the front or top of the range. Electrical controls shall be located on a set-back panel. The ranges shall be designed so that repairs or replacements can be made from the front. The ranges shall be constructed entirely of stainless steel except for the griddle top, motor and housing, other miscellaneous hardware and wiring.

3.4.1 General.

3.4.1.1 Inclined operation. When specified, for shipboard or submarine use (see 6.2), ranges shall operate satisfactorily when the base is inclined at an angle of 15-degrees each side of the vertical in each of two vertical planes at right angles to each other, with no spillage of fluid or product when tested as specified in 4.4.10.

3.4.1.2 Magnetic permeability. When specified (see 6.2), the complete range shall be constructed entirely of 300 series stainless steel specified in 3.3.2 except for griddle top (see 3.5.6) and motor, motor housing and other miscellaneous hardware. A non-magnetic material is defined as one having a magnetic permeability of less than 2.0 after fabrication, when tested as specified in 4.4.11.

3.4.2 Electrical characteristics. Ranges shall be suitable for operation on one of the following nominal system voltages, as specified (see 6.2):

440 volts A.C., 60 Hertz (Hz), 3-phase
240 volts A.C., 60 Hertz, 3-phase
208 volts A.C., 60 Hertz, 3-phase
230 volts, D.C. with 115-volts A.C. controls and motor
115 volts, D.C. with 115-volts A.C. controls and motor

3.4.2.1 Kilowatt input. Total kilowatt (kW) input to the range shall be 24 kW maximum for type I, 25.6 kW maximum for type II, and 22 kW maximum for type III ranges when tested as specified in 4.4.8.

Source: https://assist.dla.mil -- Downloaded: 202640-21T23:10Z
Check the source to verify that this is the current version before use.

3.4.2.2 Main power switch and signal lights. A main power switch to energize the heating elements and two signal lights labeled "MAIN POWER ON" and "OVEN READY" shall be provided.

3.4.2.3 Control box. For Navy shipboard or submarine installation (see 6.2), a separate control box shall be provided for 230 volt D.C. and 115 volt D.C. power source, to contain contactors, wiring etc.

3.4.3 Thermostats. Oven thermostats shall be calibrated at 400°F and griddle-plate thermostats at 350°F. Oven and griddle-plate thermostats set at their calibration points shall maintain this average temperature, plus or minus 25°F, when tested as specified in 4..4.2. The oven temperature amplitude per cycle (difference between the average maximum and average minimum temperature), shall not exceed 40°F, and the griddle plate temperature amplitude per cycle shall not exceed 70°F, when tested as specified in 4.4.2. Thermostat controls shall have an indicator light and shall be marked in increments of 25°F with a minimum range of 150°F to 450°F and an "OFF" position for the range top and a minimum range of 150° to 450°F, and an "OFF" position for the oven. The oven temperature sensing bulb shall be provided with a protective guard and covering on the sensing leads.

3.4.4 Type I, size 1 and 2 range griddles. The type I, size 1 range griddle top shall be 24 inches deep by 36 inches wide (maximum). The type I, size 2 range griddle top shall be 23 inches deep by 42 inches wide, consisting of four thermal areas and each with a thermostat control.

3.4.4.1 Type II, size 1 and 2 range griddle and hot plates. The type II, size 1 range griddle top shall be 24 inches by 24 inches and shall be controlled by two thermostats. The two hot plates shall each be controlled by an infinite heat switch or three heat, four position (high, medium, low and off) indicating switch as applicable. One thermostat shall control each 12 inch by 24 inch of griddle top section. The griddle plate shall be provided with side and rear splash guards, the hot plate control switches are to be labeled. The type II, size 2 range griddle top shall be 18 inches deep by 23 inches wide and shall be controlled by two thermostats. One thermostat shall be required to control an area of 9 inches by 23 inches of the griddle top. The griddle plate shall be provided with side and rear splash guards. The two French hot plates shall each be 9 inches diameter (minimum) and be controlled by a six heat switch (high, medium high, medium, medium low, low, warm and off). The hot plate switches are to be labeled. The 2/3 size griddle shall be controlled by two thermostats and the two French hot plates shall each be controlled by an infinite heat switch or a six heat, seven position indicating switch.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:10Z
Check the source to verify that this is the current version before use.

3.4.4.2 Type III, size 1 and 2, range with 6 hot plates. The type III, size 1 and 2 (see 3.5.2) range shall have six 9 inch diameter (minimum) French hot plates or six 8-1/2 by 9-1/2 inches (minimum regular hot plates and shall each be controlled by an infinite heat switch or a three heat (four position indicating switch (high, medium, low and off). The hot plate switches are to be labeled.

3.4.5 Style 1, fully assembled ranges. The style 1 range shall be assembled by the manufacturer, be complete and ready for installation in accordance with his standard practices.

3.4.5.1 Style 2, modular bolted construction range. The style 2 range shall be furnished as modules to be bolted together at the installation site to form a complete and usable convection oven range. Size 1 modules shall be sized to be able to pass through a 25 inch diameter opening or a door with a 20 inch wide dimension (maximum), and a height of 38 inches with 10 inch radii, top and bottom (see 6.1). All modules shall be completely wired and equipped with the necessary inter-connections by means of terminal blocks or connector plugs appropriately marked for proper connections. The range when assembled shall be of rigid and durable construction.

3.5 Construction.

3.5.1 Frame. The range frame shall be of angle or reinforced sheet metal construction with joints firmly secured and adequately support a uniform load of 75 pounds per square foot when tested as specified in 4.4.5.

3.5.2 Range size 1 and 2, exterior dimensions. The size 1 range shall be 36 inches wide, 30 inches high (excluding 6 inch legs and backsplash) and 38 inches deep (excluding grabrail). For size 2, the range shall be 42 inches wide, 46 inches high (including 5 inch base and backsplash) and 28-1/2 inches deep (excluding grabrail) (see 3.4).

3.5.3 Range oven, size 1 and 2, interior dimensions. The size 1 oven interior dimensions shall be 20 inches wide, 14 inches high (minimum) and 27 inches deep. For size 2 ranges, each oven cavity shall be 28 inches wide, by 11 inches high and 19 inches deep (minimum) (see 3.4).

3.5.4 External metal surfaces. The front sides and back of the range shall be stainless steel specified in 3.3.2, not less than 20 gauge (0.035) inches thick.

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:10Z
Check the source to verify that this is the current version before use.

3.5.5 Range tops. The portion of the range top in front of the surface heating units shall be of stainless steel as specified in 3.3.2, not less than 16 gauge (0.59) inches thick or may be nickel plated carbon steel, not less than 18 gauge (0.047) inches thick. The entire range top shall be reinforced and secured to the range body in such a manner as to insure against warping or buckling when heated. The type I top shall have two grease chutes and one for the type II top, and two spillage drawers shall be provided with each range. A back splash shall be furnished with each range and shall be fabricated from 18 gauge (0.048) inches thick (minimum) stainless steel as specified in 3.3.2 or nickel plated carbon steel 18 gauge (0.047) inches thick as specified in 3.3.4.

3.5.6 Top heating surfaces. The griddle surface shall be one piece cast iron as specified in 3.3.1, carbon steel as specified in 3.3.4, or stainless steel as specified in 3.3.2. Hot tops shall be in accordance with 3.3.1, ribbed or reinforced underneath to minimize warping with at least 3/8 inch thick ribbing. Heating surfaces shall not be porous or pitted, and shall be flat within 1/8 inch, when tested as specified in 4.4.9.

3.5.7 Heating elements. Heating elements shall be of the tubular type conforming to MIL-H-22577, and shall be securely fastened, shall not sag and shall be removable through the front of the oven. The insulation resistance of each range at normal operating temperature, shall not be less than 1 megohm. Insulation resistance at room temperature shall not be less than 2 megohms.

3.5.8 Electric wiring. All wiring shall be enclosed, adequately secured and protected against spillage and mechanical damage. Metal edges with which the wire may come in contact, shall be well rounded or covered. Wiring terminal connections shall be tight and secure. All terminal leads shall be labeled or suitably marked as to their function. A ground strap shall be provided from the range to its mounted support or located area. Wiring shall meet the requirements of NFPA 70.

3.5.9 Grab-rail. The range, when specified for shipboard or submarine (see 6.2), shall be provided with a stainless steel grabrail, full width across the front. The grabrail shall extend 3-1/2 to 4 inches beyond the front of the range, level with the top, approximately. The grabrail shall be supported by a sufficient number of brackets and hardware to insure rigidity and facilitate removal for range passage thru doors and hatches.

3.5.10 Sea-rails, type II and III ranges. Type II and III ranges, when specified for shipboard or submarine use shall be provided with sea rails, located around the hot plates to prevent from sliding off of hot plates.

3.5.11 Convection oven.

3.5.11.1 Oven door. The oven door may be either drop counter-balance type or side opening type. For all style 2, size 2 ranges when specified for shipboard or submarine use (see 6.2), the oven door shall be drop counter-balance type. The oven door shall be spring counter-balanced, and when fully opened shall lie

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:10Z
Check the source to verify that this is the current version before use.

and remain level and even with the oven bottom. The door opening shall be the full width and height of the oven interior. The oven door when open shall be capable of supporting a dead-weight load 50 pounds per square foot without breakage or permanent distortion when tested as specified in 4.4.6. The side opening type oven door when fully open shall remain in the fully open position and shall open flush with the oven interior sidewalls. The fully opened door shall not obstruct the removal of the oven shelves and, when applicable, shelf supports. Means shall be provided for lubricating door hinges, if not permanently lubricated type. Oven doors shall be equipped with a door latch. An electric interlock shall be provided to de-energize the blower and heating elements when the oven door is opened (optional for style II, size 2 range).

3.5.11.2 Oven chamber linings. Oven chamber linings shall be stainless steel, as specified in 3.3.2, or aluminized steel as specified in 3.3.3, and shall be of a welded construction without openings, except as required for blower penetration and oven vent.

3.5.11.3 Oven lining panels. Oven lining panels shall be fabricated from not less than 20 gauge (0.035) inch thick stainless steel, as specified in 3.3.2, and shall be used in conjunction with the oven chamber linings (panels optional for style II, size 2 ranges). Oven lining panels shall be easily cleanable and shall be removable through the oven door without the use of tools.

3.5.11.4 Oven shelves. The oven compartment shall be furnished with six removable shelves with positive stops. The six shelf spacing shall be 2 inches minimum. The shelf shall be fabricated of steel wire rod, steel frame with bar grating, or expanded steel metal of stainless as specified in 3.3.2, or steel as specified in 3.3.4, chrome plated to the manufacturer's standard practice. Each shelf shall be capable of supporting a load of 20 pounds per square foot without more than 1/8 inch deflection or permanent deformation when tested as specified in 4.4.7. Each loaded shelf shall not bind when placed into, or removed from, the shelf supports when tested as specified in 4.4.7.

3.5.11.5 Oven blower. The oven blower shall provide even circulation and heat distribution throughout the oven. The blower shall be powered by a 1725 rpm ball-bearing motor not smaller than 1/3 horsepower for the voltages as specified (see 3.4.2 and 6.2).

3.5.11.6 Oven heat loss. The energy required to maintain the oven at set temperature (oven heat loss), shall not be greater than 59 watts per square foot of interior oven area when tested as specified in 4.4.4.

3.5.11.7 Oven preheat time. The oven temperature shall rise 330°F in not more than 20 minutes, when tested as specified in 4.4.3.

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:10Z
Check the source to verify that this is the current version before use.

3.6 Range hold-down means. When specified for shipboard use (see 6.2), four 6 inch high stainless steel hold-down legs shall be securely bolted to the bottom corners of the range body, or a 5 inch high frame (minimum), open to the front, shall be bolted in place. A hole shall be provided for a 1/2 inch diameter bolt (minimum) in each leg. Hold down bolts shall be accessible from the front of the range.

3.7 Marking. Identification and instructions shall be permanently and legibly marked directly on the item or on an aluminum, brass, or stainless steel plate firmly affixed to the item. Marking shall be stamped, embossed, engraved or applied by photosensitive means. The plate thickness shall be not less than 0.012 inch.

3.7.1 Identification. Each range identification plate shall include the manufacturer's model and serial number, and name, trade name or trademark of such known character as to be readily identifiable. In addition, such information as required by UL 197, shall be put on the identification plate. The instruction plate shall be located on the outside front of the oven door.

3.8 Finish. Range finishes shall be free from discoloration and stains. Stainless steel shall be finished in accordance with manufacturer's standard practice.

3.9 Workmanship. All components and assemblies of the range shall be free from dirt and other extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings and molded parts if used, shall be free of sand, flashing, pits, blowholes, and sprues. External surfaces shall be free from sharp edges and sharp corners.

3.9.1 Metal fabrication. Metal used in the fabrication of the range shall be free from kinks. Forming and shearing shall not cause damage to the metal and the metal shall be free from trim and tool marks.

3.9.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be smooth and free from cracks, burn holes, undercuts and incomplete fusion. All scale and flux shall be removed from the weld area.

3.9.3 Fastening devices. Holes punched or drilled shall be free of burrs. Threaded fasteners shall not be broken, cracked or stripped and shall be drawn tight. Rivets, when used, shall fill the hole completely and the heads shall be in full contact with the surface of the member.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:10Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.2.2 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of end items. The sample unit shall be one end item. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not as specified	X	
	Discoloration or stains present	X	
	Stainless steel not a smooth finish	X	
Construction and workmanship (general)	Component missing	X	
	Component fractured, buckled, kinked, bent, dented, punctured, malformed	X	
	Any burr, sliver, sharp edge; rough die, tool, gouge, trim, or grind marks		X

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:10Z
Check the source to verify that this is the current version before use.

TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Top heating surfaces	Pitted, porous, cracked, dented, chipped rusted	X	
	Part missing	X	
	Adjustable part of assembly that cannot be adjusted	X	
	Part misplaced, loose, or not in proper alignment		X
Body	Not constructed as specified	X	
	Materials not as specified	X	
Oven	Open joints in linings	X	
	Linings not reinforced or supported	X	
	Oven lining panels or racks not removable through oven door	X	
	Dampered vent not furnished	X	
	Heating elements not as specified	x	
Shelves	Not removable	X	
	Not number specified	X	
Heating elements	Not as specified	X	
	Not securely fastened; sagging	X	
	Not removable from front of oven	X	
	Sag	X	
Oven blower	Not accessible for cleaning or servicing	X	
Indicator lights	Not provided as specified	X	
Thermostats and switches	Not as specified	X	
Top heating surfaces	Not of materials specified	X	
	Back-splash not furnished	X	
	Grease-chutes and spillage drawers not furnished	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:10Z
Check the source to verify that this is the current version before use.

TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Identification plates	Missing, not of specified materials	X	
	Not marked as specified (on item or on plate)	X	
	Incomplete, illegible	X	
Shipboard use	Range, not suitable for marine duty, as specified	X	

4.2.3 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of end items. The sample unit shall be one end item. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.2.4 End item testing.

4.2.4.1 Performance testing. The initial production end item or a representative end item selected at random from and existing production lot shall be tested as specified in 4.4.1 through 4.4.11. Any failure of the initial production end item shall be cause for rejection of the item, for necessary design and construction changes, and for repetition of the above testing on the next produced end item. The process shall be repeated until an end item successfully passes the specified tests. Any failure of an end item selected from and existing production lot shall be cause for rejection of that lot.

4.2.4.2 Operational testing. Every end item shall be tested as specified in 4.4.1. Any end item exhibiting one or more test failures shall be rejected.

4.2.4.3 Load, door, and shelf testing. The end item s shall be tested as specified in 4.4.5, 4.4.6, and 4.4.7. The lot size shall be expressed in units of end items. The sample unit shall be one end item. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0 for test failures.

4.2.5 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of MIL-R-11337.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:10Z
Check the source to verify that this is the current version before use.

4.3 Certification compliance. Certificates of compliance, certified test reports, or listing marks for codes and standards, as applicable, that one submitted as proof of compliance with document requirements, shall be examined and validated.

4.4 Methods of inspection. Tests when specified in 4.4.1 through 4.4.4 shall be conducted in an temperature ambient of 75°F, plus or minus 5°F.

4.4.1 Operation test. The empty range shall be connected to a source of power having the same electrical characteristics specified in the contract or order. All controls. including switches and thermostats, and all controlled components, such as heating elements, shall be operated to insure that they function as specified in 3.4, 3.4.2.2, 3.4.3.3, 3.4.4, 3.4.4.1, 3.4.4.2 and 3.5.11.1. Any nonconformance shall constitute failure of this test.

4.4.2 Thermostat test. The range shall be connected to the required source. The oven temperature shall be measured by a thermocouple suspended from or supported by an oven rack so that the thermocouple junction is at least 2 inches from any metal part, is in the center of the usable baking space, and is shielded from direct radiation. Surface temperatures for types I and II griddle top and types II and III hot plate(s) shall be determined by use of an indicating surface pyrometer of thermocouple secured to the surface being measured. The griddle top surface temperature readings shall be taken at 5 locations per heat zone, one each 3 inches from the corners and one located at the point of intersection of the diagonals. The hotplate surface temperature reading shall be taken at the center area of each hotplate. If a thermocouple is used, the hot junction shall be brazed or welded. With the applicable thermostat set at its calibration point, the oven (later, the top heating surface) shall be operated until the steady-heat condition is reached. Steady-heat condition is attained after the heating elements have been under the control of the thermostat for at least the first five cycles of operation of the thermostat, as shown by the indicator light. When this condition is reached, the maximum and minimum oven and top heating surface temperatures shall be taken for three consecutive cycles and the average calculated for compliance with 3.4.3. The oven temperature amplitude per cycle, and the top plate temperature amplitude per cycle, shall be noted. Any nonconformance with the requirements specified shall constitute failure of this test.

4.4.3 Oven heating-time test. The oven shall be at ambient temperature with the door closed before starting the test. The thermostat shall be set at 450°F and the convection blower energized. The temperature shall be read at 1 minute intervals. The time required to raise the oven temperature 330°F shall be recorded to determine compliance with 3.5.11.7. Any nonconformance shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:10Z
Check the source to verify that this is the current version before use.

4.4.4 Oven heat-loss test. Install a watt-hour meter in the main electrical line to the range (A voltmeter and ammeter installed in the line along with a clock or stop watch may be used to compute the watt hours used, in place of the watt hour meter. Volts and amps shall be recorded a minimum of four times during the test, with the average volt and ampere readings as recorded to determine the wattage used). Locate a thermocouple in the center of the oven cavity. Set the thermostat at 330°F above the ambient temperature. One oven only is to be energized for this test. Oven is to be operated for a minimum of one hour at the specified set temperature for conditioning prior to starting the test. The test time shall be for one hour and the watt hour meter (or alternate meters as specified), shall be read at the beginning and at the end of the one hour test period. The total watts as required to maintain the steady internal oven temperature, for one hour, shall be determined and the average watts per square foot of oven interior surface area, shall be computed to determine conformance to 3.5.11.6.

4.4.5 Load test. The top of the assembled range shall be uniformly loaded with 75 pounds per square foot. After 15 minutes the load shall be removed and the range examined for compliance with the structural strength requirements of 3.5.1. Any nonconformance shall constitute failure of this test.

4.4.6 Door structural strength. With the oven door in the open position a uniformly distributed load of 50 pounds per square foot shall be applied to the top surface. After 5 minutes the load shall be removed and the door examined to determine compliance with requirements specified in 3.5.11.1. Any nonconformance shall constitute failure of this test.

4.4.7 Shelf structural strength. The shelf shall be placed at its normal position in the oven and subjected to a uniformly-distributed load of 20 pounds per square foot. After 2 minutes the load shall be removed and the shelf examined for compliance with the shelf-strength requirements of 3.5.11.4. After deflection test has been completed, withdraw loaded shelf to stop position and return to normal shelf position to determine compliance with shelf binding requirements specified. Any nonconformance shall constitute failure of this test.

4.4.8 Kilowatt input test. With the range connected to the specified power source and all heating elements energized on "HIGH", the total power input shall be measured for compliance with 3.4.2.1. Any nonconformance shall constitute failure of this test.

4.4.9 Range-top flatness test. A straightedge shall be placed over both diagonals of the types I and II griddle-tops, over types II and III hot plate(s), and over the combined hot-top section. Examination shall be made for compliance with the flatness requirements of 3.5.6. Any nonconformance shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:10Z
Check the source to verify that this is the current version before use.

4.4.10 Inclined operation test. For this test the grease drawer(s) shall be filled to 30 percent capacity. Operation the range for 5 minutes with the base inclined at an angle of 15 degrees from each side of the vertical in each of two vertical planes at right angles to each other. At each of these positions observe for conformance with specified requirements of 3.4.1.1. Any non-conformance with specified requirements shall constitute failure of this test.

4.4.11 Permeability test. Magnetic permeability shall be tested in accordance with ASTM A 342 to determine conformance with 3.4.1.2. Any nonconformance shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. The range oven equipment shall be preserved in accordance with the applicable requirements of MIL-R-11337.

5.1.2 Commercial. The range oven equipment shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A and B packing. The range oven equipment shall be packed in accordance with the applicable requirements of MIL-R-11337, preserved as specified in 5.1.

5.2.2 Commercial packing. The range oven equipment shall be packed in accordance with ASTM D 3951, preserved as specified in 5.1.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-R-11337 or ASTM D 3951, as applicable.

5.4 Use of polystyrene (loose-fill) material. Unless otherwise approved by the acquisition activity(see 6.2), use of polystyrene (loose-fill) material for packaging applications such as cushioning, filler and dunnage is prohibited. When approved, unit packs and containers (interior and exterior) shall be marked and labeled as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:10Z
Check the source to verify that this is the current version before use.

"CAUTION

Contents cushioned, etc., with polystyrene (loose-fill) material shall not be taken aboard ship.
Remove and discard loose-fill material before shipboard storage of equipment.
If required, recushion with cellulosic material bound fiber, fiberboard or transparent flexible cellular material."

6. NOTES

6.1 Intended use. The ranges are intended for high speed griddle cooking and forced-air convection-oven cooking, in kitchens and galleys on Navy ships and submarines.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Type, grade, size and style as required.
c. When required for shipboard submarine or Naval shore installations use (see 3.4.1.1, 3.4.2.3, 3.5.9, 3.5.10, 3.5.11.1, and 3.6).
d. When a non-magnetic range is required (see 3.4.1.2).
e. Electrical characteristics (see 3.4.2 and 3.5.11.5).
f. Selections of applicable levels of preservation, and packing (see 5.1 and 5.2).

6.3 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD

Preparing activity:

Army - GL

Project No. 7310-0698

Review activities:

Army - MD
Navy - SA

Source: https://assist.dla.mil -- Downloaded: 202610-21T23:10Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY
US Army Natick Research and Development Center
Natick, Massachusetts 01760

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
STRNC-ES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research and Development Center
ATTN: STRNC-ES
Natick, MA 01760

Source: https://assist.dla.mil -- Downloaded: 202640-21T23:10Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-43954A	Ranges, Convection Oven, Electric

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 1426 82 MAR — PREVIOUS EDITION IS OBSOLETE.

NATICS OP, 1 Apr 84

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T23:10Z
Check the source to verify that this is the current version before use.